

Year 7 • Lesson 10.03 Dot plots: answers (teacher copy)

10.03 Dot plots (booklet p. 3)

Warm-up

- 3, 3, 5, 7, 9
- 3
- $\frac{1}{5}$
- 21

WE DO examples

- Ex 1: 5 students have 2 pets; 20 students surveyed
- Ex 2: mode 1; outlier 7
- Ex 3: dots at 12 (3), 13 (1), 14 (2), 15 (1), 20 (1)
- Ex 4: mode 12; cluster from 12 to 15; outlier 20
- Ex 5: scale 0 to 10; dots at 2 (1), 5 (1), 6 (3), 7 (6), 8 (5), 9 (3), 10 (1)
- Ex 6: most scores cluster from 6 to 9 (mode 7); 2 is an outlier, so one student may need help

Set A

- 20
- 6
- 4
- 8 hours
- 6 hours
- 12 hours
- 12
- 10
- 4
- 3

Set B

- Frequencies: size 6; 2, 7: 4, 8: 6, 9: 3, 10: 1
- Dot plot matching the table
- 8
- 3
- 4
- 6
- 6
- No
- $\frac{6}{16} = \frac{3}{8}$
- 10

Set C

- 26 °C
- 34 °C
- 6
- 5
- $\frac{3}{30} = \frac{1}{10}$
- $\frac{6}{30} = 20\%$
- 10
- $\frac{7}{30}$
- About 24 °C to 27 °C (most days 23 °C to 29 °C)
- e.g. a heatwave or hot westerly wind on one day

Set D

- Scale 0 to 30 (or 3 to 30); dots: 3 (1), 4 (2), 5 (7), 6 (6), 7 (4), 8 (2), 9 (1), 12 (1), 30 (1)
- 5
- 30 (12 is also well away from the cluster)
- 5
- $\frac{10}{25} = \frac{2}{5}$
- $\frac{6}{25} = 24\%$
- Most values cluster from 4 to 8
- No. The mode is still 5 because it has the most dots
- Small data set of whole numbers, so every value can be shown
- e.g. Most students sent 4 to 8 texts, 5 was most common, and one student sent 30

Extension

- Any 10 values with the most at 3 and one far away
- 10 sevens, so 5 eights
- Too many values and heights have decimals, so the dots would not fit; use a stem-and-leaf plot or grouped graph

Exit ticket

- Level 1: 4
- Level 2: $\frac{4}{12} = \frac{1}{3}$
- Level 3: outlier 9; the other values cluster from 2 to 5 (mostly 3 to 4)